

DAY 2: INTEGRATION MASTERY - PROBLEMS

9 PHYSICS PROBLEMS

Instructions

Solve all 10 problems showing complete work. All questions are physics-based and require integration techniques. Include units in your answers. Box your final answers.

Problem 1: Motion with Variable Acceleration

The acceleration of a particle moving along a straight line is given by $a(t) = 12t^2 - 6t$ m/s². At $t = 0$, the particle is at rest at the origin.

- Find the velocity function $v(t)$.
- Find the position function $x(t)$.
- What are the velocity and position at $t = 2$ seconds?
- Find the displacement of the particle between $t = 0$ and $t = 3$ seconds.

Problem 2: Distance vs. Displacement

A particle moves with velocity $v(t) = t^2 - 6t + 8$ m/s for $0 \leq t \leq 6$ seconds.

- Find when the particle changes direction (solve $v(t) = 0$).
- Find the displacement (net change in position) over the interval.
- Find the total distance traveled.
- Sketch the velocity vs. time graph and shade the area representing distance.

Problem 3: Work Done by a Spring

A spring exerts a force $F(x) = -kx$, where $k = 200$ N/m and x is the displacement from equilibrium in meters.

- How much work is required to stretch the spring from $x = 0$ to $x = 0.25$ m?
- How much work is required to stretch it from $x = 0.1$ m to $x = 0.3$ m?
- If the spring is compressed from $x = 0$ to $x = -0.2$ m, how much work is done by the spring?
- A 0.5 kg mass is attached to the spring and released from rest at $x = 0.2$ m. What is its speed when it passes through $x = 0$? (Use work-energy theorem)

Problem 4: Work to Lift a Chain

A 15-meter long chain has mass 30 kg and hangs vertically from one end. ($g = 9.8$ m/s²)

- What is the linear density λ of the chain in kg/m?
- Consider a small link of the chain of length dy at a distance y meters from the top. What is the mass of this link? How much work is required to lift this link to the top?
- Set up and evaluate the integral to find the total work required to lift the entire chain to the top.
- How much work is required to lift only the bottom half of the chain to the top?

Problem 5: Work to Pump Water

A cylindrical tank of radius 2 m and height 8 m is filled with water. The tank is oriented vertically. (Water density $\rho = 1000$ kg/m³, $g = 9.8$ m/s²)

- Consider a thin horizontal slice of water at depth y meters from the top, with thickness dy . What is the volume of this slice? What is its mass?
- How much work is required to pump this slice to the top of the tank?

- c. Set up and evaluate the integral to find the total work required to pump all the water to the top.
- d. How much work is required if the water must be pumped to a point 3 meters above the top of the tank?

Problem 6: Variable Density Rod

A rod of length $L = 4$ m lies along the x -axis from $x = 0$ to $x = 4$. Its linear density varies as $\lambda(x) = 3 + 2x$ kg/m.

- a. Find the total mass of the rod.
- b. Find the center of mass x_{cm} of the rod.
- c. Find the moment of inertia about an axis perpendicular to the rod through the left end ($x = 0$).
- d. Use the parallel axis theorem to find the moment of inertia about an axis through the center of mass.

Problem 7: Moment of Inertia of a Rod with Exponential Density

A rod of length $L = 2$ m has density $\lambda(x) = 5e^{-x/2}$ kg/m, where x is measured from the left end.

- a. Find the total mass M of the rod.
- b. Find the center of mass x_{cm} .
- c. Find the moment of inertia about an axis perpendicular to the rod through the left end.
- d. Is the moment of inertia larger or smaller than that of a uniform rod of the same mass and length? Why?

Problem 8: Hydrostatic Force on a Dam

A dam has a vertical face that is shaped like a trapezoid. The bottom is 20 m wide, the top is 40 m wide, and the height is 15 m. The water level is at the top of the dam. (Water density $\rho = 1000 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$)

- Place the y -axis with $y = 0$ at the water surface, increasing downward. Find an expression for the width $w(y)$ of the dam at depth y . (Use similar triangles)
- For a thin horizontal strip at depth y with thickness dy , what is the area of the strip? What is the pressure at this depth?
- What is the force on this strip?
- Set up and evaluate the integral to find the total force on the dam.
- If the water level drops by 3 meters, how does the force change?

Bonus Challenge Problem

Problem 9: Moment of Inertia of a Cone

A solid cone of height $H = 6 \text{ m}$ and base radius $R = 2 \text{ m}$ has uniform density $\rho = 500 \text{ kg/m}^3$. The cone's axis is along the z -axis, with vertex at the origin and base at $z = H$.

- Find the radius as a function of height z .
- Find the mass of the cone using a single integral (slicing perpendicular to the z -axis).
- Find the moment of inertia about the z -axis. (Use cylindrical coordinates: each disk has $I = \frac{1}{2}r^2 dm$)
- If the cone rotates about its axis at 10 rad/s , what is its rotational kinetic energy?